

EXPERIMENT-1

DATE-

Objective: Prepare a SRS document in line with the IEEE recommended standards.

Description: The SRS for the MRI Brain Tumor Detection System using Machine Learning provides a clear description of all requirements needed to develop an automated tool for detecting and classifying brain tumors from MRI images. It defines the purpose, scope, users, system functions, and constraints according to IEEE standards. The document outlines how the system will upload MRI scans, preprocess images, identify tumor regions, classify tumor types, and generate diagnostic reports. It also specifies performance expectations, security needs, hardware/software requirements, and system limitations. Overall, the SRS serves as a complete blueprint that guides the design, development, and evaluation of the machine learning-based tumor detection system.

Solution:

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define and describe the functional and non-functional requirements of the MRI Brain Tumor Detection System using Machine Learning. This document serves as a reference for developers, testers, project managers, and stakeholders to ensure a clear understanding of system capabilities, constraints, and design expectations.

1.2 Scope

The system aims to automatically detect and classify brain tumors from MRI scans using machine learning algorithms such as CNN-based deep learning models. The software processes MRI images, performs preprocessing, extracts features, trains a model, and predicts tumor presence, type, and affected region. The system will be used by radiologists, researchers, and medical professionals to support faster and more accurate diagnosis.

1.3 Definitions, Acronyms, and Abbreviations

- **MRI** – Magnetic Resonance Imaging
- **ML** – Machine Learning

- **CNN** – Convolutional Neural Network
- **ROI** – Region of Interest
- **GUI** – Graphical User Interface
- **TP/TN** – True Positive / True Negative
- **FP/FN** – False Positive / False Negative

1.4 References

- IEEE 830-1998 SRS Guidelines
- IEEE 29148-2018 Software & Systems Requirements
- Research papers on MRI-based brain tumor detection
- Medical imaging datasets (BraTS, Kaggle MRI dataset)

1.5 Overview

The SRS document contains detailed descriptions of system functionality, performance constraints, software interfaces, user characteristics, system features, and security requirements. It also includes assumptions and dependencies relevant to the project.

2. Overall Description

2.1 Product Perspective

The MRI Brain Tumor Detection System is an independent system that can be integrated with existing hospital management or radiology systems. It digitizes the tumor detection process with automated image processing and ML-based prediction.

2.2 Product Functions

The system will:

- Allow uploading MRI scans in DICOM/JPG/PNG formats
- Perform preprocessing (noise removal, normalization)
- Segment the brain region
- Detect and classify tumor types (e.g., Glioma, Meningioma, Pituitary)
- Highlight tumor region using bounding boxes/heatmaps

- Generate a prediction report
- Provide accuracy metrics for model performance

2.3 User Classes and Characteristics

- **Radiologists:** expert users, require accurate and clear outputs
- **Medical Students/Researchers:** moderate technical understanding
- **Technicians:** basic operational use (uploading images, viewing reports)

2.4 Operating Environment

- **Hardware:** Desktop/Laptop with GPU (optional), 8–16 GB RAM
- **Software:** Windows/Linux, Python 3.x, TensorFlow/PyTorch, Flask/Django for UI
- **Tools:** OpenCV, NumPy, Scikit-learn

2.5 Design and Implementation Constraints

- Limited availability of labeled medical datasets
- High-resolution MRI images require strong computation
- Models must comply with medical ethics and privacy standards
- The system cannot replace final diagnosis by radiologists

2.6 User Documentation

- User manual
- Installation guide
- Model training and evaluation guide

2.7 Assumptions and Dependencies

- MRI images provided are of clinical quality
- Dataset is pre-labeled for initial training
- The user has basic computer knowledge
- Internet may be required for cloud-based processing

3. Specific Requirements

3.1 Functional Requirements

FR1: Image Upload

- The system shall allow users to upload MRI images in DICOM/JPG/PNG format.
- The system shall validate image size and format.

FR2: Image Preprocessing

- The system shall normalize MRI images.
- The system shall apply noise reduction techniques.
- The system shall perform skull stripping and segmentation.

FR3: Tumor Detection & Classification

- The system shall use a ML model to detect tumor presence.
- The system shall classify tumor type (Glioma/Meningioma/Pituitary).
- The system shall locate tumor region using heatmaps or bounding boxes.

FR4: Result Generation

- The system shall display the predicted class with probability score.
- The system shall generate a PDF/HTML report summarizing detection results.
- The system shall show segmented tumor regions.

FR5: Model Training (Admin Only)

- The system shall allow retraining using new datasets.
- The system shall display accuracy, loss, sensitivity, specificity.

FR6: User Interface

- The system shall provide a GUI for uploading images and viewing results.
- The interface shall be user-friendly and responsive.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

- The system shall process each MRI scan within **5–10 seconds** (GPU-enabled).
- Accuracy should be **≥ 90%** for tumor detection.
- The system shall support high-resolution MRI images up to **512×512** pixels.

3.2.2 Safety Requirements

- Misclassification risks must be clearly communicated.
- The tool shall not replace a radiologist's final diagnosis.

3.2.3 Security Requirements

- The system shall secure MRI images using encryption.
- The system shall restrict model retraining to authenticated users.
- Patient data shall not be stored without consent.

3.2.4 Software Quality Attributes

- **Reliability:** 99% uptime, consistent predictions.
- **Usability:** Simple interface with step-by-step workflow.
- **Maintainability:** Modular code structure for easy updates.
- **Portability:** Compatible with Windows and Linux.

4. External Interface Requirements

4.1 User Interface

- Dashboard page
- Image upload page
- Results viewer with segmentation display
- Training dashboard (admin)

4.2 Hardware Interface

- GPU (optional) for faster computation

- MRI data input from local storage

4.3 Software Interface

- TensorFlow/PyTorch backend
- OpenCV for image processing
- Database for storing logs and results

4.4 Communication Interfaces

- HTTP/HTTPS for web-based interaction
- REST API calls for model inference (if deployed online)

5. System Features Summary

- Automatic brain tumor detection
- MRI image preprocessing pipeline
- Deep learning-based classification
- Tumor region highlighting
- User-friendly prediction report
- Secure and scalable system

6. Other Requirements

- Compliance with medical data privacy standards (HIPAA-like principles)
- Future scalability for multi-class tumor detection
- Ability to integrate with PACS/Radiology systems

7. Appendices

- Appendix A: Sample MRI dataset description
- Appendix B: Model architecture diagrams
- Appendix C: Glossary of medical and ML terms

EXPERIMENT-2

DATE-

Objective: Draw the use case diagram and specify the role of each of the actors.

Description: The use case diagram for the MRI Brain Tumor Detection System using Machine Learning illustrates how different users interact with the system to complete their respective tasks. The system supports operations such as uploading MRI scans, processing and detecting tumors, displaying results, generating reports, and retraining the machine learning model. Each actor in the diagram performs a clearly defined role, ensuring the smooth operation and usability of the system.

Solution:

1. Radiologist – (Primary Medical User)

Specific Role:

The Radiologist uses the system to upload MRI images, view tumor detection results, analyze the highlighted tumor region, and use the generated report to support medical diagnosis. Their main responsibility is to interpret the final output for clinical decision-making.

2. Technician – (Operational Support User)

Specific Role:

The Technician assists in the technical handling of MRI images. They upload and manage MRI scans in the system, ensuring correct file formats and image quality. They do not perform diagnosis but help keep the workflow smooth.

3. Researcher – (Analytical User)

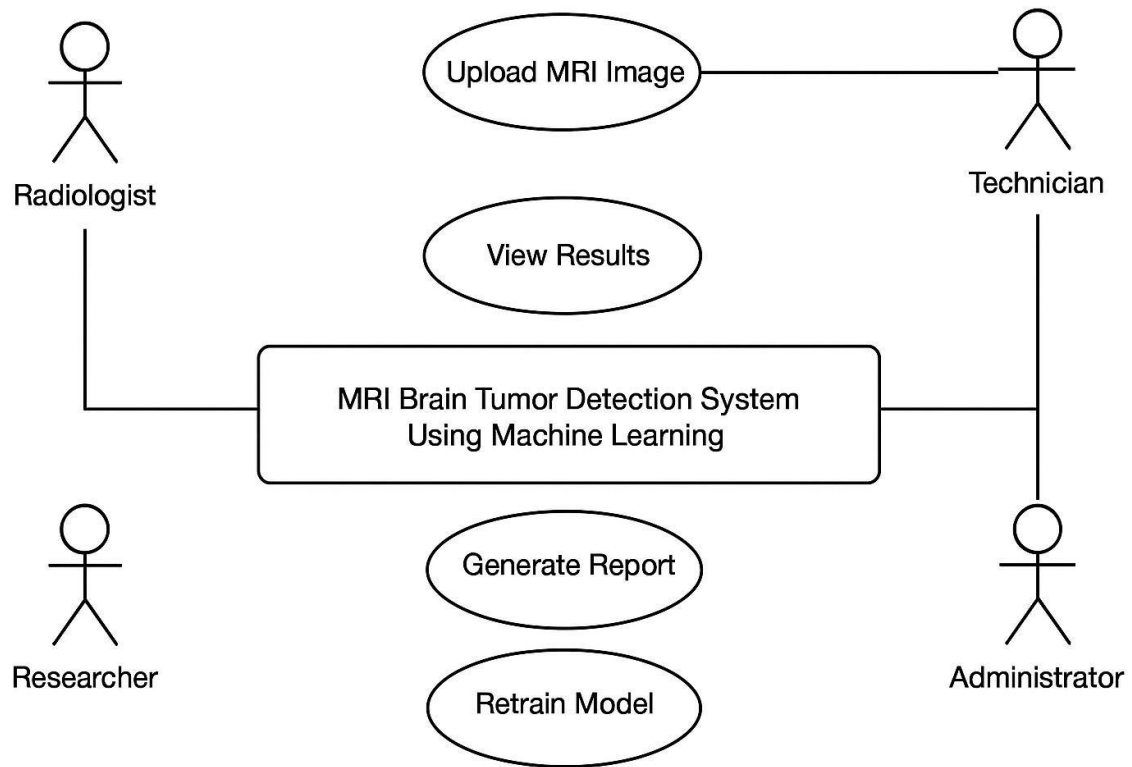
Specific Role:

The Researcher uses the system to study detection patterns, validate results, and generate reports for academic or clinical research. They may compare model outputs or use data to evaluate system effectiveness.

4. Administrator – (System Manager)

Specific Role:

The Administrator controls system settings, manages user access, retrains the machine learning model with new datasets, monitors system performance, and ensures accurate and secure operation of the entire application.



The Use Case Diagram

EXPERIMENT-3

DATE-

Objective: Prepare state the precondition, post condition and function of each use case.

Description:

1. User Login

Allows authorized users to access the system securely by verifying their login credentials.

2. Upload MRI Image

User uploads the MRI scan to the system for further processing and tumor analysis.

3. Preprocess Image

System enhances and cleans the uploaded MRI image to prepare it for machine learning-based detection.

4. Detect Tumor

The machine learning model analyzes the preprocessed MRI and determines whether a tumor is present.

5. Segment Tumor Region

System highlights and extracts the tumor area from the MRI image for better visualization.

6. View Result Report

Displays the detection output, tumor segmentation, confidence score, and related findings to the user.

7. Download Report

Allows the user to download the complete diagnosis report in PDF format for offline use.

8. Manage User Account (Admin)

Admin manages system users by adding, modifying, or removing user accounts.

Solution:

1. Use Case: User Login

Short Description

Allows authorized users (doctor/admin) to access the system securely.

Precondition

- User must be registered.
- System must be connected to the database.

Postcondition

- User Dashboard loads after successful login.

Function

- Validates username and password.
- Grants access to system functionalities.

2. Use Case: Upload MRI Image

Short Description

User uploads brain MRI images for processing.

Precondition

- User must be logged in.
- MRI image must be in acceptable format (JPG, PNG, DICOM).

Postcondition

- Image stored in system.
- Image ready for preprocessing.

Function

- Accept MRI file.
- Validate and store for further processing.

3. Use Case: Preprocess Image

Short Description

Removes noise and enhances MRI for accurate tumor detection.

Precondition

- MRI image must be uploaded successfully.

Postcondition

- Preprocessed image generated and saved.

Function

- Apply filters, resizing, normalization.
- Prepare image for ML model.

4. Use Case: Detect Tumor (ML Model Execution)

Short Description

System detects the presence of a tumor using a ML model.

Precondition

- Preprocessed image must be available.
- ML model must be loaded successfully.

Postcondition

- Tumor detection result stored (Yes/No).
- Probability/scores generated.

Function

- Run trained ML model on MRI.
- Produce detection output.

5. Use Case: Segment Tumor Region

Short Description

Highlights and extracts only the tumor area in the MRI image.

Precondition

- Tumor detected or image preprocessed.
- Model must support segmentation.

Postcondition

- Segmented image created and stored.

Function

- Mask tumor area.
- Generate segmented output for visualization.

6. Use Case: View Result Report

Short Description

Shows tumor detection, segmentation, and additional model outputs.

Precondition

- Detection/segmentation must be completed.

Postcondition

- Report displayed on screen.

Function

- Present detection status.
- Display images, confidence scores, and tumor size (if applicable).

7. Use Case: Download Report

Short Description

User downloads the analysis report in PDF.

Precondition

- Report must be generated successfully.

Postcondition

- Report saved on the user's device.

Function

- Convert analysis results into PDF.
- Enable user to save the report.

8. Use Case: Manage User Account (Admin)**Short Description**

Admin manages user profiles such as adding, removing, or updating users.

Precondition

- Admin must be logged in.

Postcondition

- User data updated in the system.

Function

- Add users.
- Update roles.
- Remove accounts.

EXPERIMENT-4

DATE-

Objective: Draw the activity diagram.

Description: The activity diagram for MRI Brain Tumor Detection illustrates the entire process followed by the system. The flow begins with the user uploading an MRI image. The system then preprocesses the image to remove noise and enhance quality. Next, the machine learning model checks whether a tumor is present.

If a tumor is detected, the system performs tumor segmentation to highlight the affected region. Whether a tumor is found or not, the system finally displays the detection result to the user, and the process ends. The diagram clearly shows the decision-making step and the different paths the system takes based on the tumor detection outcome.

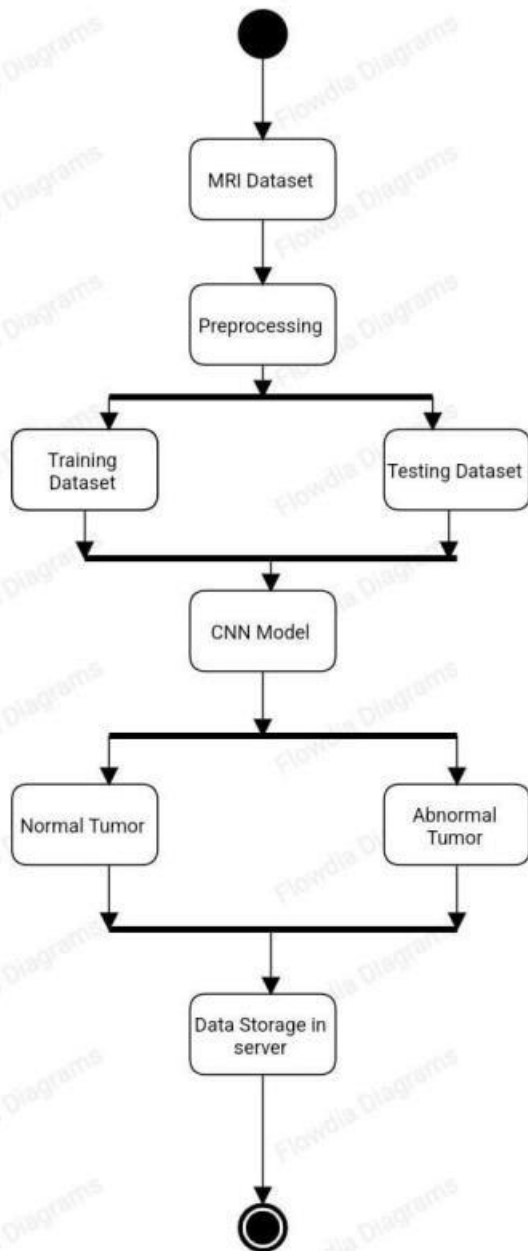
Solution: An Activity Diagram is a behavioral UML diagram used to model the workflow of a system.

It shows the sequence of activities, decision points, parallel operations, and how the system moves from one activity to another.

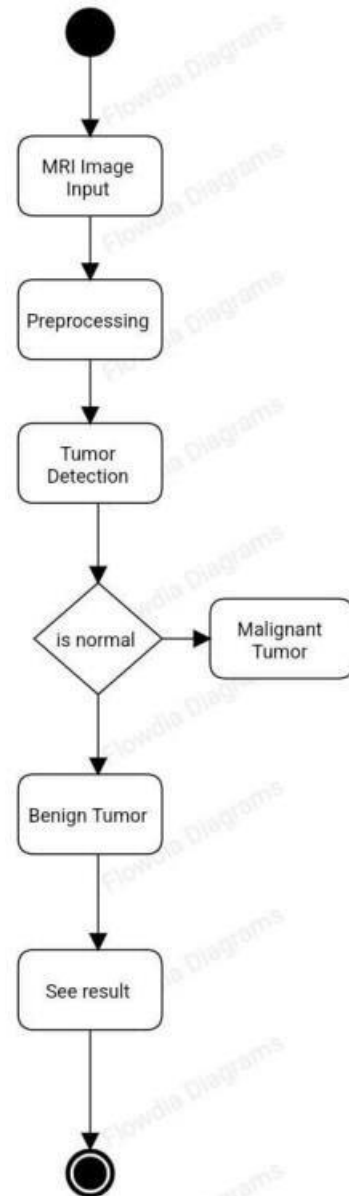
In this project, the activity diagram explains the step-by-step flow of the MRI tumor detection system — from uploading an image to viewing the final result. It helps in understanding how the system processes the input image, applies machine learning algorithms, and generates the tumor detection report.

Key purposes of an activity diagram:

- Represents the flow of control in the system
- Describes business logic and processing steps
- Shows decision points (Yes/No conditions)
- Useful for visualizing system behavior before implementation



Activity Diagram for System



Activity Diagram for Students

EXPERIMENT-5

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Objective: Identify the classes. Classify them as weak and strong classes and draw the class diagram.

Description: The classes in the MRI Brain Tumor Detection System are identified based on the core functions — uploading MRI images, processing them, applying ML algorithms, and generating reports. Strong classes represent fundamental entities like the MRI image, ML model, preprocessing module, and report. Weak classes serve as helper components that support the primary workflow by storing metadata, user session details, and model performance statistics.

Solution:

Strong Classes (Main System Classes)

1. User

- Stores user information and roles.
- Handles login and authentication.

2. MRIImage

- Represents the uploaded MRI brain scan.
- Stores file details and format.

3. PreprocessingModule

- Performs noise removal, normalization, and image enhancement.

4. MLModel

- Represents the trained machine learning model.
- Performs tumor detection and classification.

5. DetectionResult

- Stores output of tumor detection.
- Contains tumor status and confidence score.

6. SegmentationModule

- Extracts and highlights the tumor region in MRI.

7. Report

- Generates and stores diagnostic report in PDF or text format.

Weak Classes (Supporting Classes)

1. LoginSession

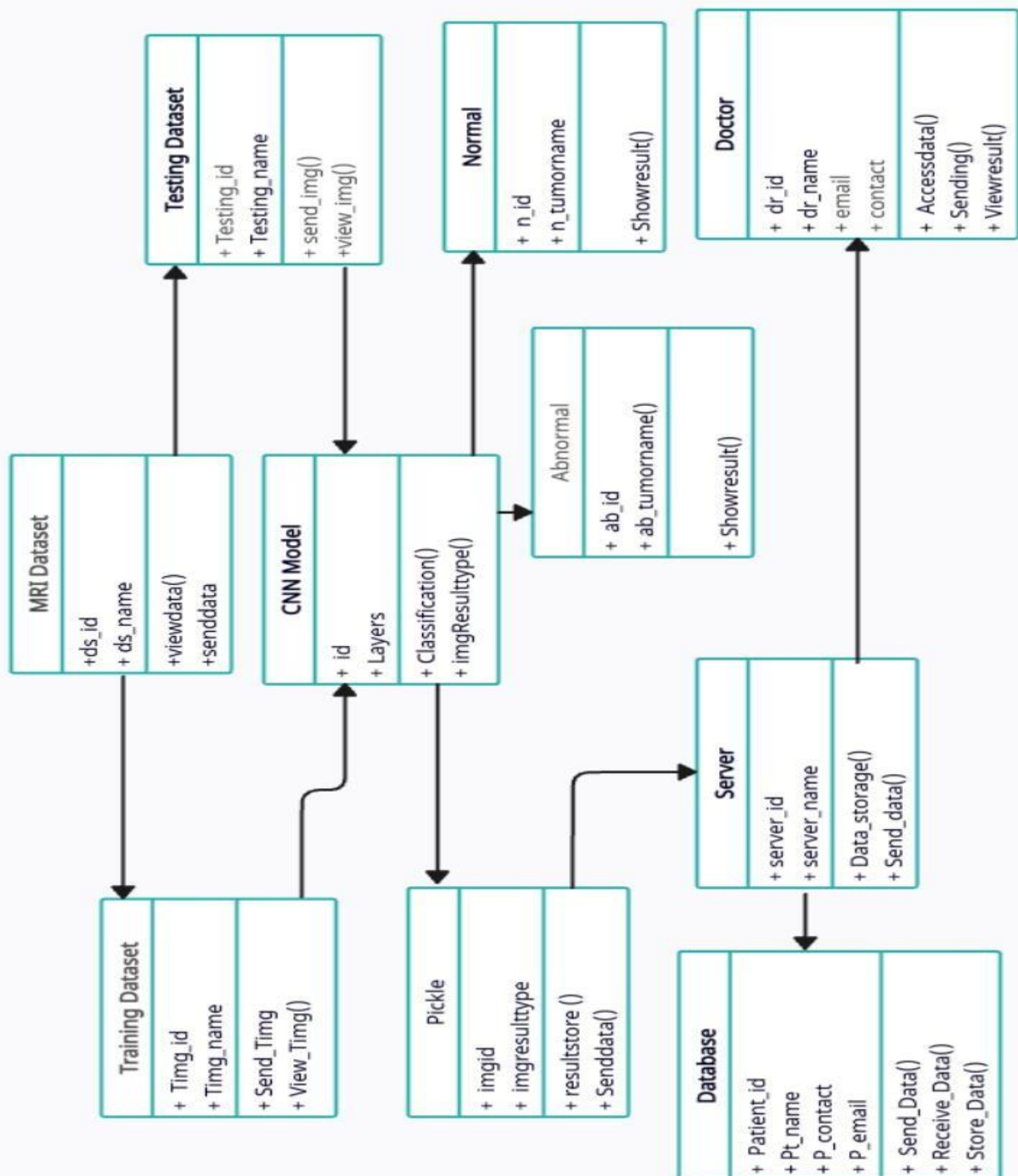
- Stores temporary user session details like login/logout time.

2. ImageMetadata

- Stores extra image information such as resolution and file size.

3. PerformanceMetrics

- Stores ML model accuracy, precision, recall, and loss values.



The Class Diagram

EXPERIMENT-6

DATE-

Objective: Draw the sequence diagram for any two scenarios.

Description:

Solution: A Sequence Diagram is a behavioral UML diagram that illustrates how objects in a system interact with each other in a time-ordered sequence. It focuses on the message flow between objects to complete a specific scenario or use case.

Sequence diagrams show:

- The objects/actors involved
- The order of messages exchanged
- The lifecycle of objects during the interaction
- How the system responds to events step-by-step

Sequence diagrams are especially useful in understanding the dynamic behavior of the system and clarifying how different modules collaborate to perform a task.

Sequence Diagram – Scenario 1: Upload MRI Image

Theory Content (Short Description)

This sequence diagram describes how a user uploads an MRI image to the system. The user initiates the upload request, the system validates the image format, stores it in the database, and confirms a successful upload.

Flow Summary

1. User selects an MRI file and clicks upload.
2. System validates file format (PNG/JPG/DICOM).
3. System stores the image in database/storage.
4. System returns a confirmation message to the user.

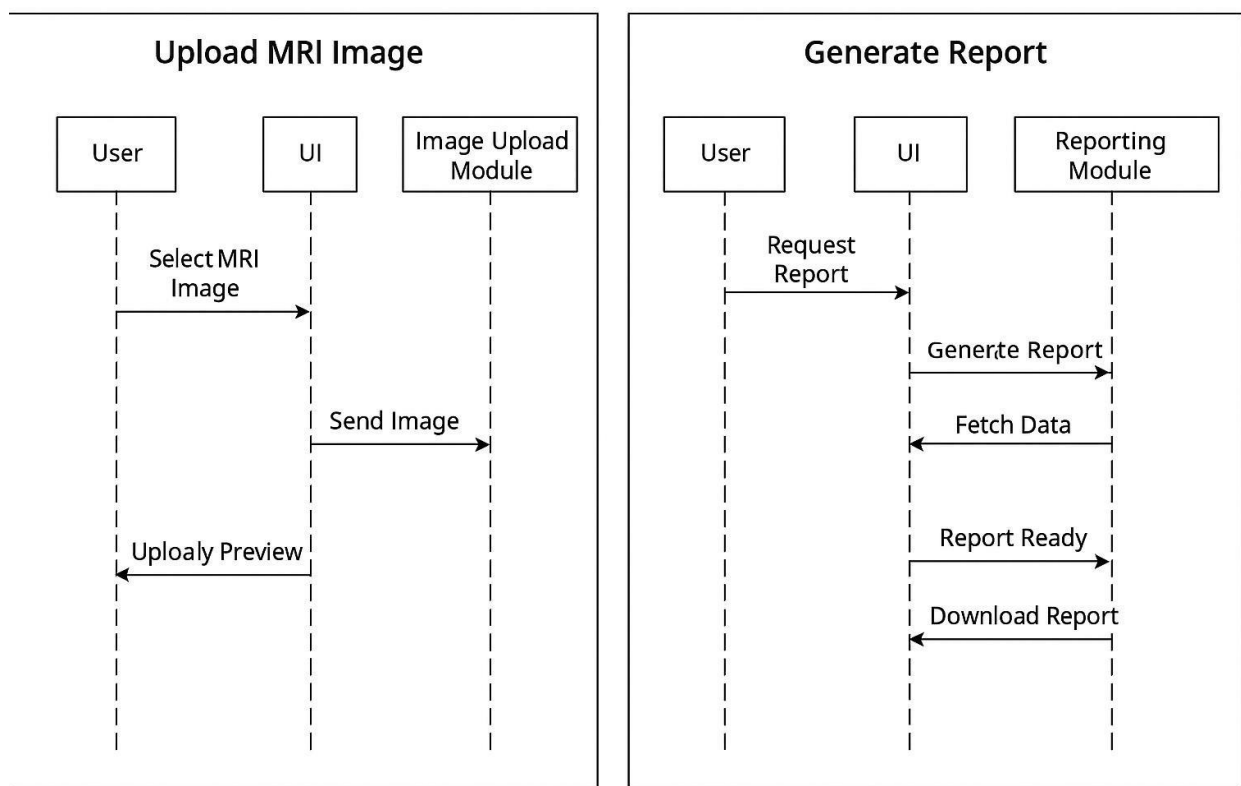
Sequence Diagram – Scenario 2: Detect Tumor Using ML Model

Theory Content (Short Description)

This sequence diagram describes how the system processes the MRI image to detect a brain tumor using a machine learning model. The system preprocesses the image, sends it to the ML model, receives the prediction, and displays the result.

Flow Summary

1. User triggers “Detect Tumor.”
2. System loads the preprocessed MRI image.
3. System runs the image through the ML model.
4. ML model returns detection status and confidence.
5. System displays tumor detection result to the user.



The Sequence Diagram

EXPERIMENT-7

DATE-

Objective: Draw the collaboration diagram.

Description: The Collaboration Diagram shows how different components of the MRI brain tumor detection system interact with each other through message exchanges. It highlights the structural organization of the system by representing objects such as the Image Input Module, Preprocessing Module, Segmentation Module, Feature Extraction Module, Machine Learning Classifier, and Result Display Module.

The user uploads an MRI image, which flows through preprocessing, segmentation, and feature extraction. The extracted features are sent to the machine learning model for tumor prediction. The prediction results are formatted and displayed to the user, and optionally stored in the database. Each message in the diagram is numbered to represent the order of interactions.

Solution:

A **Collaboration Diagram**, also known as a **Communication Diagram**, is a type of UML interaction diagram. It focuses on **how objects interact with each other** through the exchange of messages. While Sequence Diagrams focus on *time ordering*, Collaboration Diagrams emphasize the **structural organization and relationships** among system components.

In the context of **MRI Brain Tumor Detection using Machine Learning**, the collaboration diagram describes how different modules cooperate to detect and classify tumors from MRI images.

Purpose of the Collaboration Diagram

- To show **interaction between system components**
- To highlight **the flow of messages** during tumor detection
- To illustrate **module responsibilities**
- To provide a clear representation of **system architecture communication**

Objects/Components Involved

1. User
2. Image Input Module
3. Preprocessing Module
4. Segmentation Module
5. Feature Extraction Module
6. Machine Learning Classifier
7. Result Generation Module
8. Result Display Module
9. Database (optional)

Interaction Description

1. User → Image Input Module

The user uploads an MRI brain scan image.

2. Image Input Module → Preprocessing Module

The system passes the raw MRI image for cleaning and normalizing.

3. Preprocessing Module → Segmentation Module

After noise removal and enhancement, the image is forwarded to segmentation.

4. Segmentation Module → Feature Extraction Module

The tumor region is identified and sent for feature extraction.

5. Feature Extraction Module → ML Classifier

Features such as shape, texture, and intensity values are extracted.

6. ML Classifier → Result Generation Module

The classifier predicts tumor presence, type, and probability.

7. Result Generation Module → Result Display Module

Results are formatted into a readable report.

8. Result Display Module → User

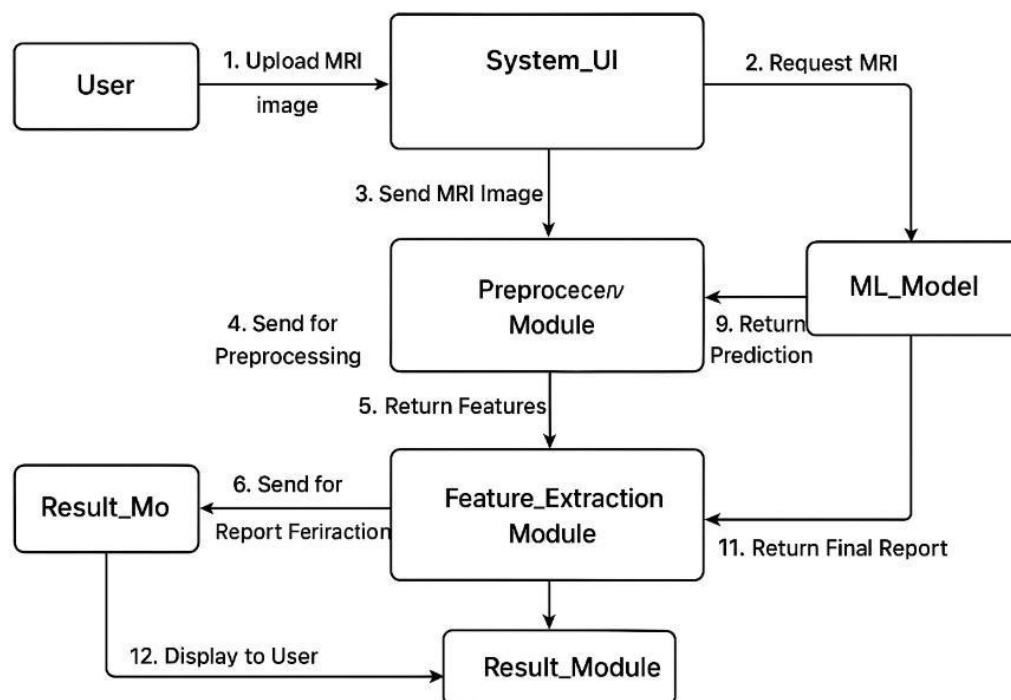
Final detection output is shown to the user.

9. Result Generation Module → Database (Optional)

System may store results, images, and logs.

- Key Characteristics of Collaboration Diagram

- Shows **structural view** of interacting objects
- Represents **numbered messages** to indicate sequence
- Emphasizes **relationships** and **communication paths**
- Useful for understanding **data flow between modules**



EXPERIMENT-8

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Objective: Draw the state chart diagram.

Description: The State Chart Diagram shows the different states the system passes through while processing an MRI scan. It starts with the image upload, then moves to preprocessing, segmentation, feature extraction, and classification. After the ML model predicts the tumor, the system generates and displays the result to the user. Each transition represents a step in the tumor detection workflow, from input image to final output.

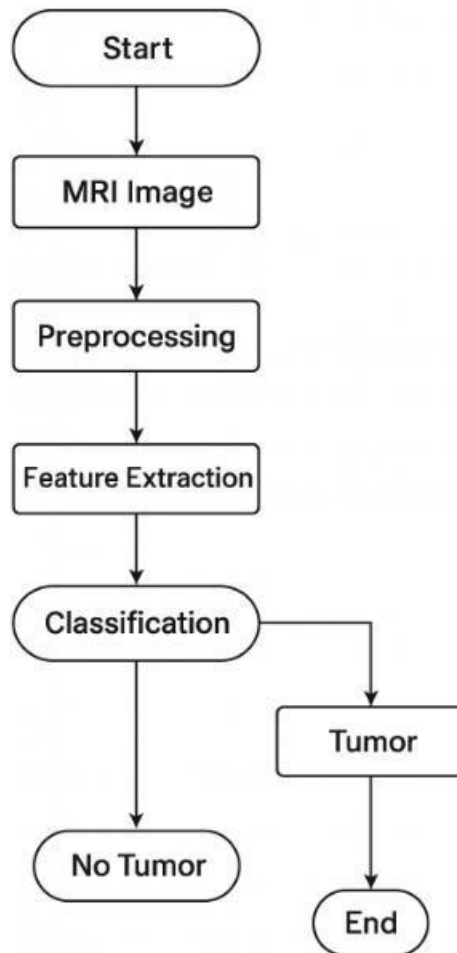
Solution:

A State Chart Diagram (State Machine Diagram) represents the different states that the system or an object goes through during its operation. It focuses on how the system transitions from one state to another based on events and conditions. In an MRI brain tumor detection system, the state chart diagram describes the sequence of functional stages involved in processing an MRI scan and predicting tumor presence.

The process begins when the user uploads an MRI image, moving the system into the *Image Upload State*. Once the image is successfully provided, the system transitions to the *Preprocessing State*, where noise is removed and the image is normalized. After preprocessing, the system enters the *Segmentation State* to identify the tumor region.

Next, the system moves to the *Feature Extraction State*, where important characteristics like texture, shape, and intensity are extracted. After features are generated, the system transitions to the *Classification State*, where the machine learning model predicts whether a tumor is present and determines its type. The system then enters the *Result Generation* state to format the prediction, and finally, the *Result Display State*, where the output is shown to the user. Once the result is delivered, the system reaches the *End State*.

The State Chart Diagram clearly shows the step-by-step flow, system behavior, and state transitions, making it useful for understanding the internal working of the tumor detection system.



The State Chart diagram

EXPERIMENT-9

DATE-

Objective: Draw the component diagram.

Description: The Component Diagram shows the major software modules of the MRI tumor detection system and how they interact. It includes components such as image input, preprocessing, segmentation, feature extraction, machine learning model, result generation, and database. The diagram highlights how each component works together in a modular structure to process MRI images and produce tumor detection results.

Solution:

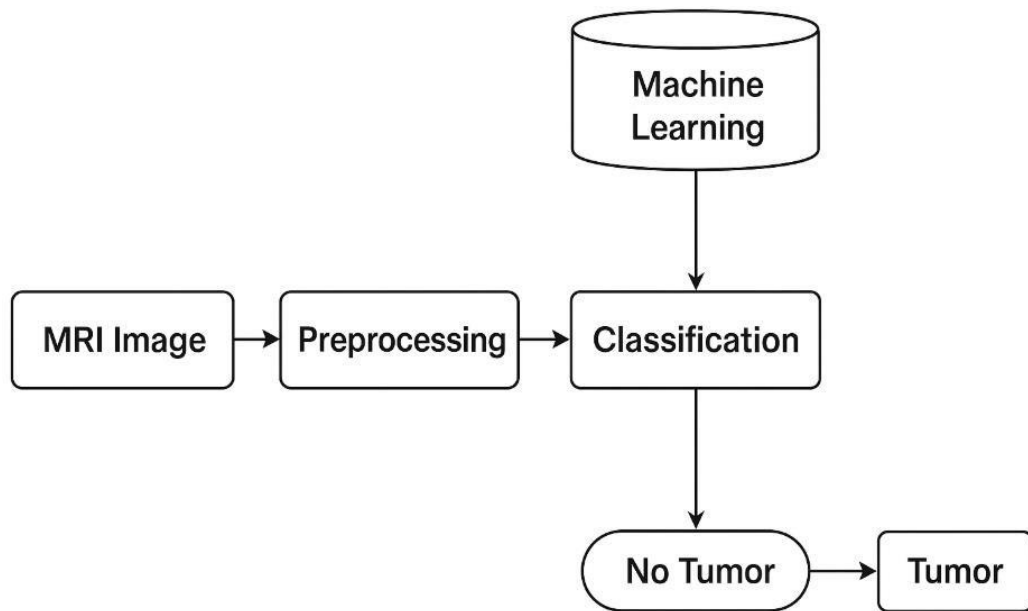
A **Component Diagram** represents the high-level structure of the system by showing its major software components and how they interact. It provides a modular view of the system architecture, illustrating how different parts of the MRI tumor detection system are organized and interconnected.

In this system, the **User Interface Component** allows the user to upload MRI images and view results. The **Image Input Component** handles image loading and validation, ensuring that the uploaded file is in the correct format. The **Preprocessing Component** removes noise, normalizes the image, and prepares it for further analysis.

After preprocessing, the **Segmentation Component** identifies the tumor region in the MRI scan. The segmented region is passed to the **Feature Extraction Component**, which extracts important features such as texture, intensity, and shape. These features are then sent to the **Machine Learning Model Component**, which performs classification to detect the presence and type of tumor.

The **Result Generator Component** compiles the prediction and produces the final diagnostic report. The system may also use a **Database Component** to store MRI images, patient details, and prediction results for future reference.

The component diagram thus highlights the modular structure of the system and shows how data flows between the components to achieve tumor detection.



The Component Diagram

Objective: Draw the deployment diagram.

Description: The Deployment Diagram shows how the MRI tumor detection system is physically deployed across different hardware nodes. It includes a client device where the user uploads MRI images, an application server that performs preprocessing, segmentation, feature extraction, and machine learning prediction, and a database server that stores images and results. It explains how these nodes communicate to deliver accurate tumor detection.

Solution:

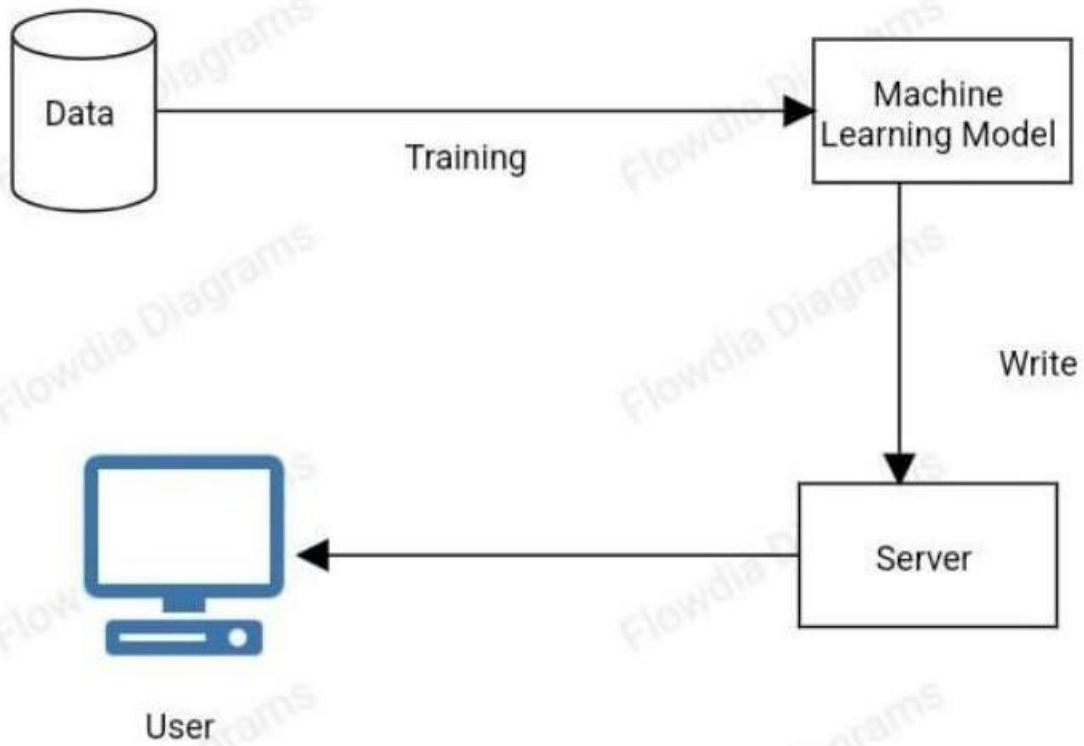
A **Deployment Diagram** is a UML diagram that represents the physical architecture of a system. It shows how software components are deployed on hardware nodes and how these nodes communicate with each other. In an MRI brain tumor detection system, the deployment diagram illustrates how different parts of the application run across client devices, servers, and databases.

The system typically includes three main nodes. The **Client Machine** (such as a laptop, desktop, or mobile device) is responsible for interacting with the user through the graphical interface. Here, the user uploads MRI images and views the final detection results. The client communicates with the backend using secure internet protocols.

The **Application Server** is the central processing unit of the system. It hosts the main software components, including image preprocessing, segmentation, feature extraction, the machine learning model, and the result generator. This server may run in a cloud or GPU-enabled environment to handle deep learning operations and intensive image computations. It processes the MRI image and generates the tumor prediction outcome.

The **Database Server** stores important data such as MRI scans, patient details, model logs, and prediction results. It maintains persistent storage and provides data to the application server whenever required. Communication between the application server and database server takes place through standard database query protocols.

The deployment diagram therefore provides a clear view of the **physical distribution**, **deployment environment**, and **inter-node communication** of the system, helping developers understand how the system is deployed in a real-world setting.



The Deployment Diagram